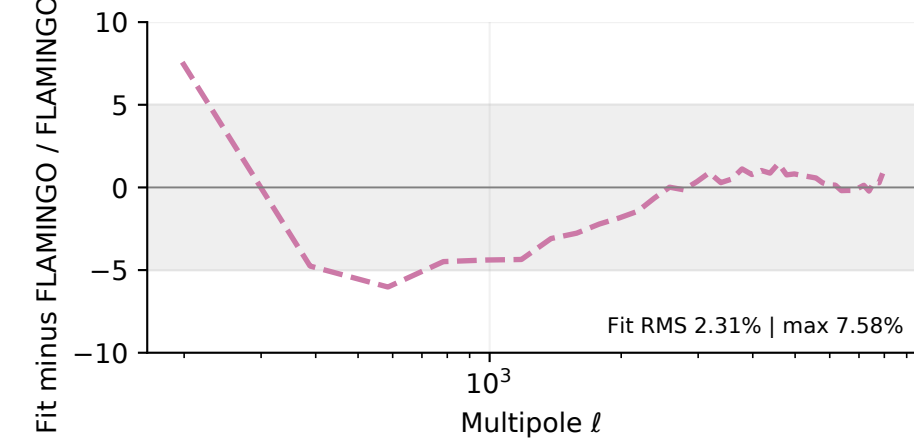
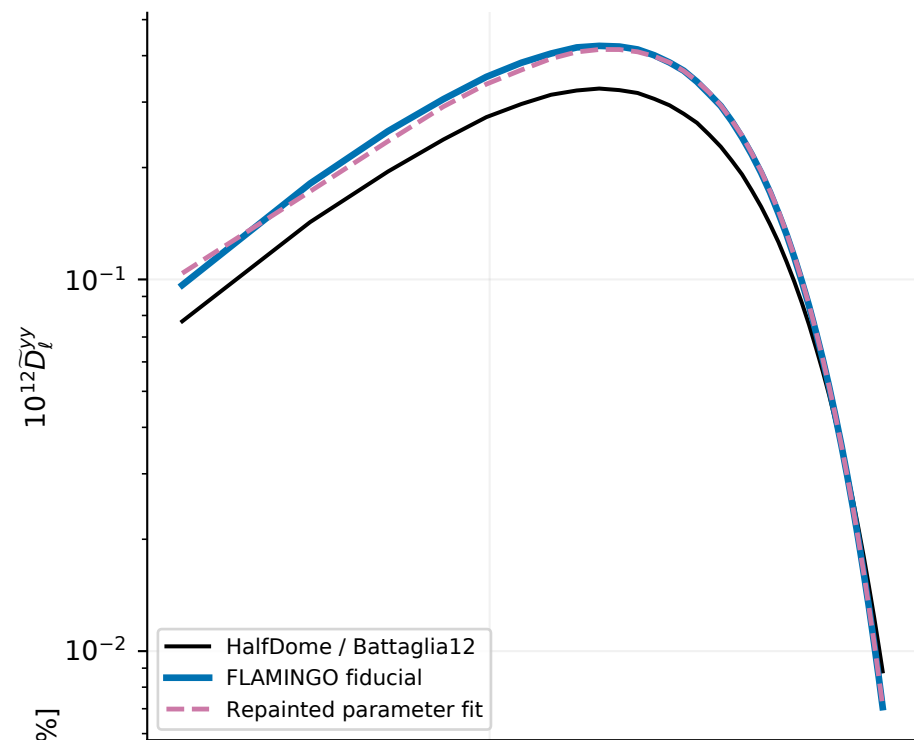
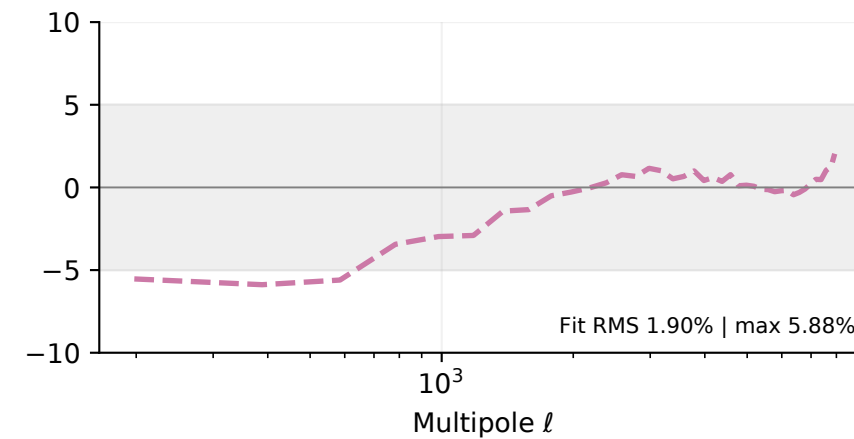
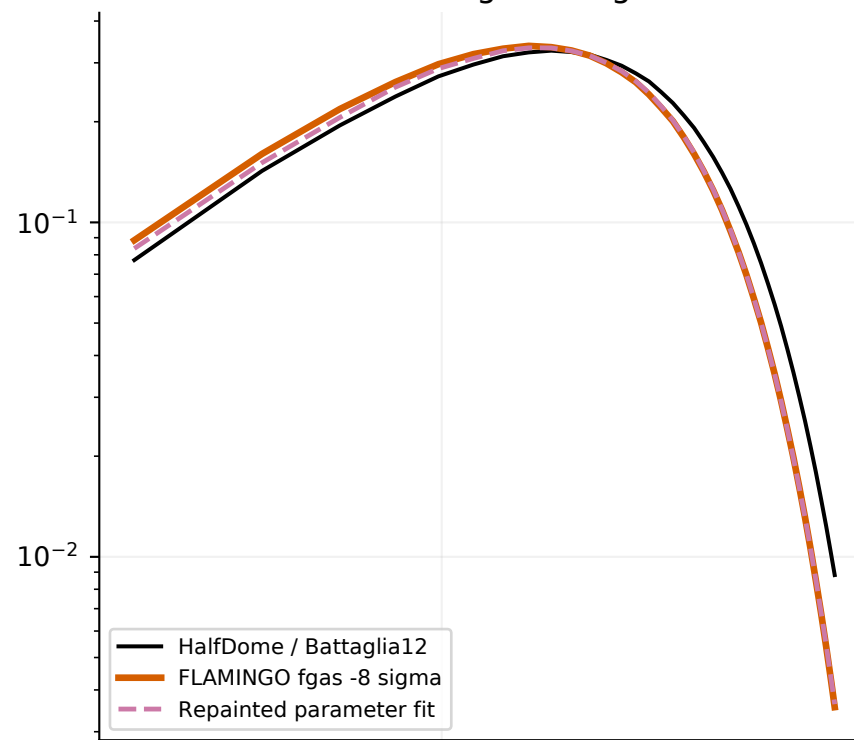


Clean spectra: full Nside 4096 maps, original 2 arcmin beam and fsky=0.4 mask
 40 bins; effective spectral matches on the HalfDome catalogue; residual shading: +/-5% bin goal

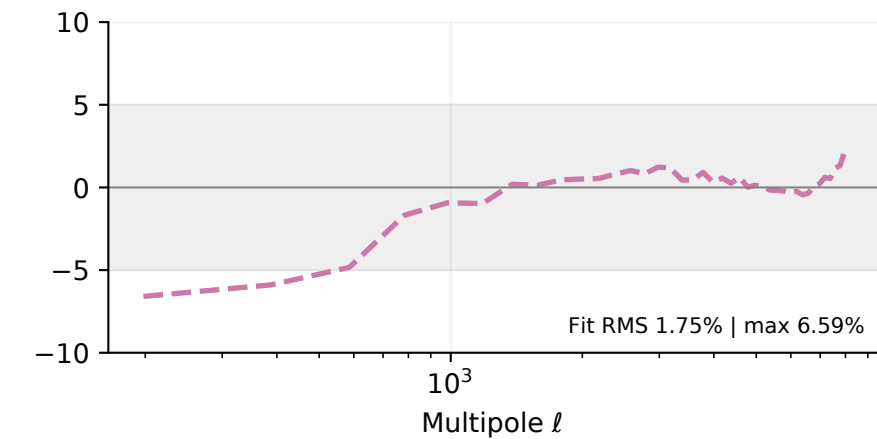
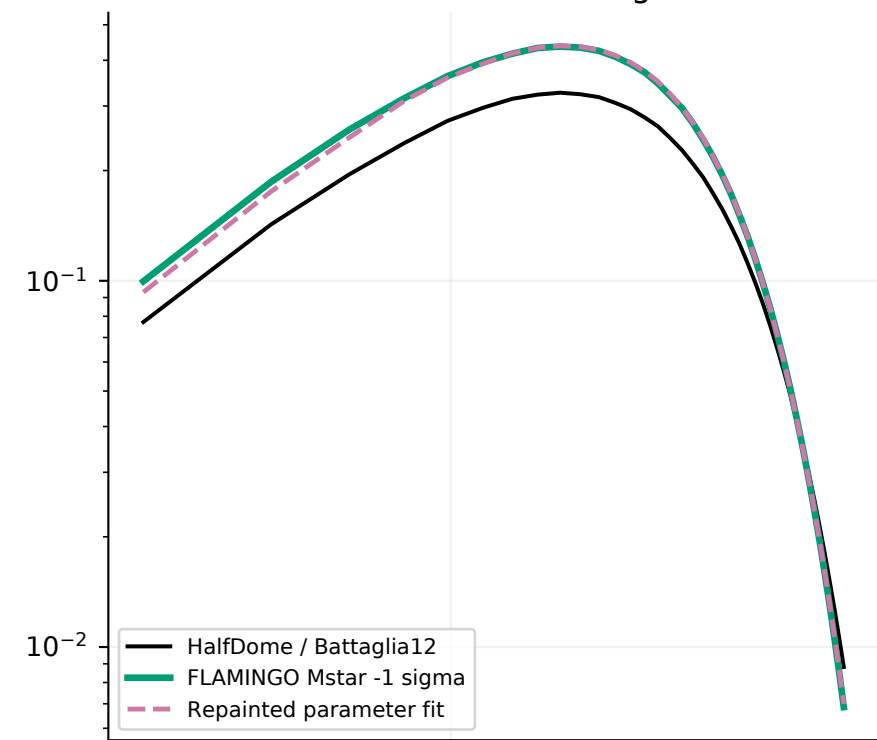
FLAMINGO fiducial



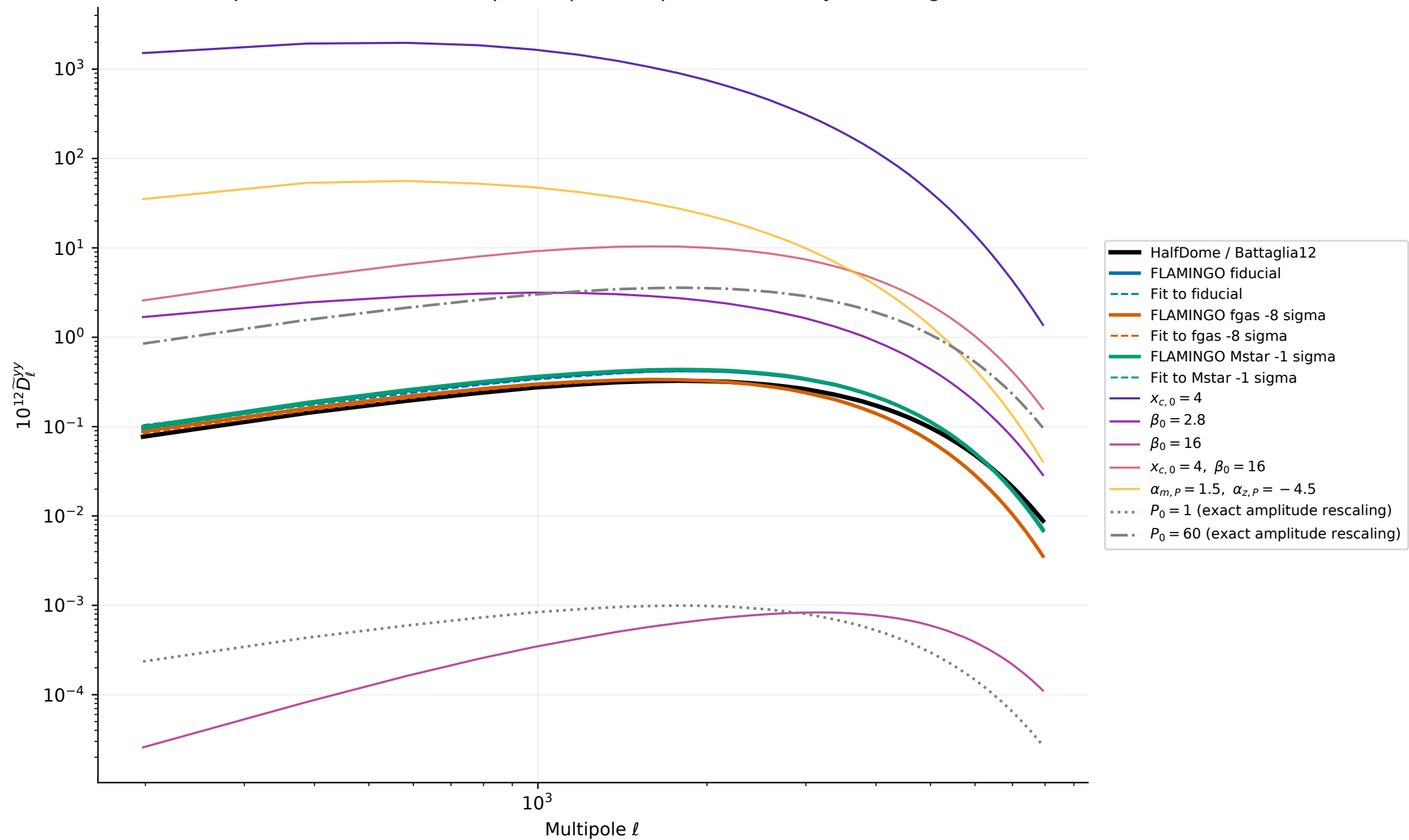
FLAMINGO fgas -8 sigma



FLAMINGO Mstar -1 sigma

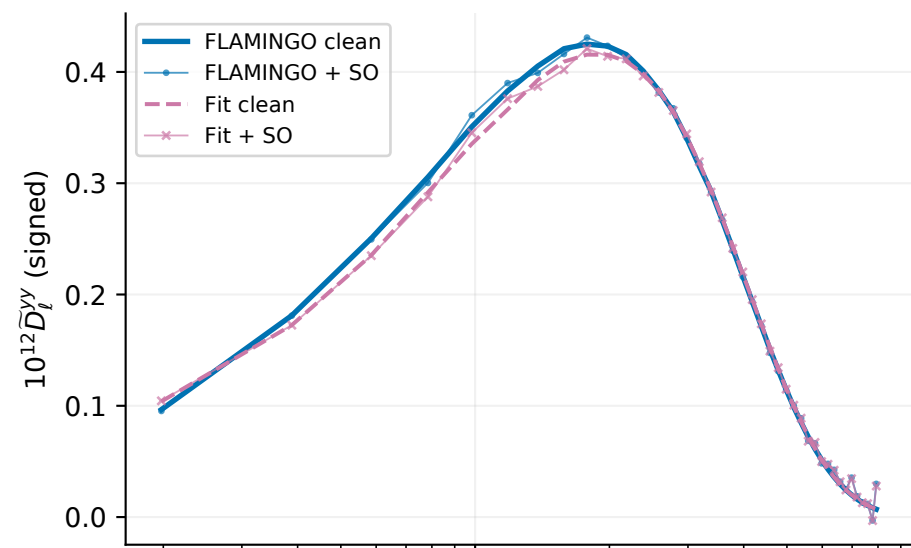


tSZ power across the proposed range
Shape extremes use full maps; unspecified parameters stay at Battaglia12

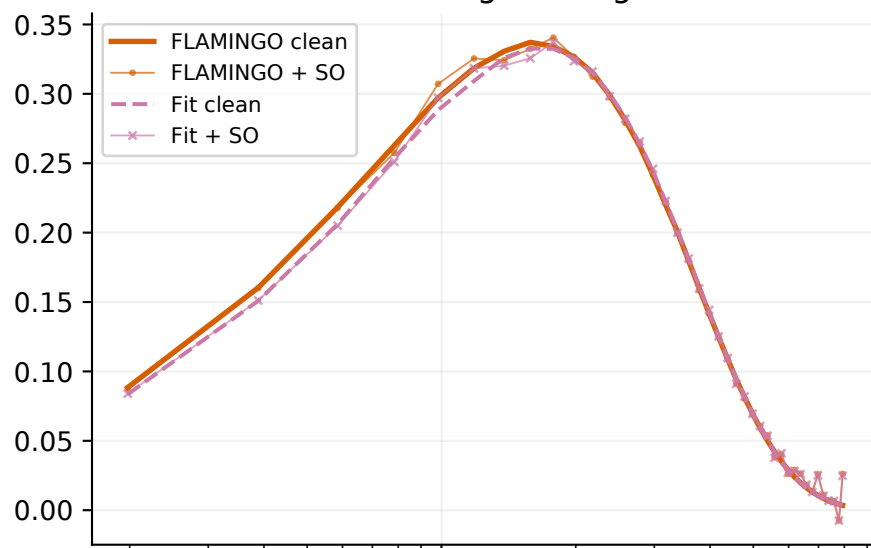


Noise shown separately: identical cached SO splits and mask for every map
 Root seed 12345; split seeds 22446 and 22447; noise was excluded from fitting

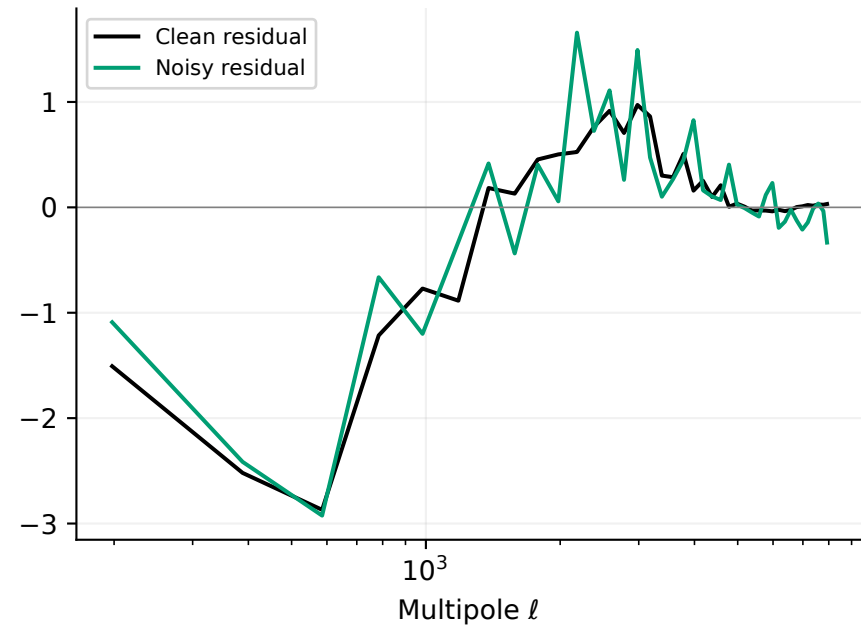
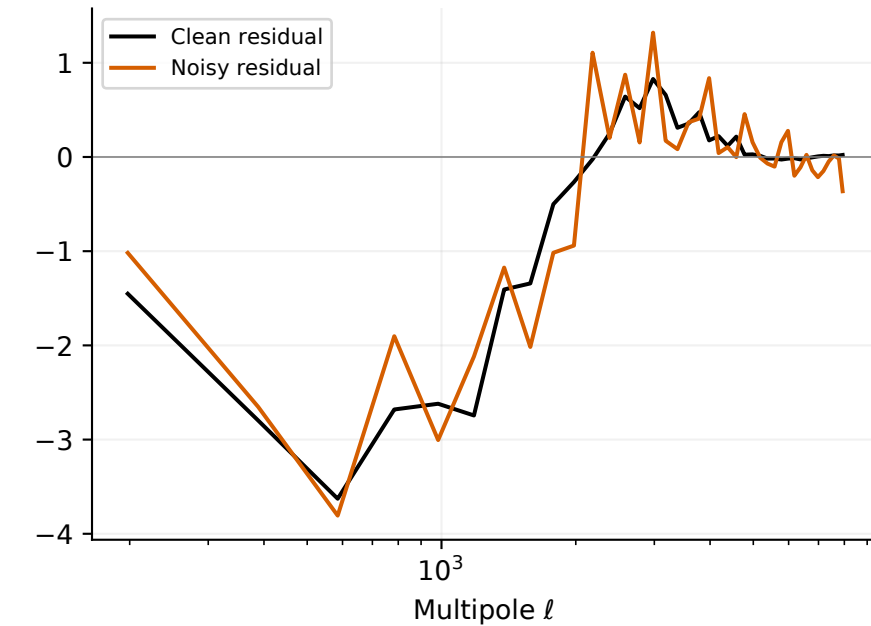
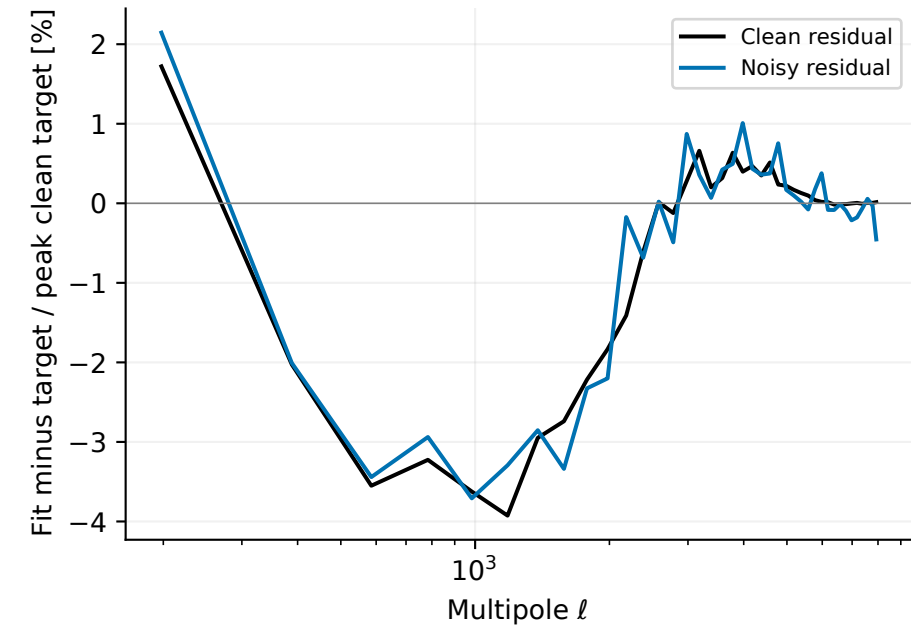
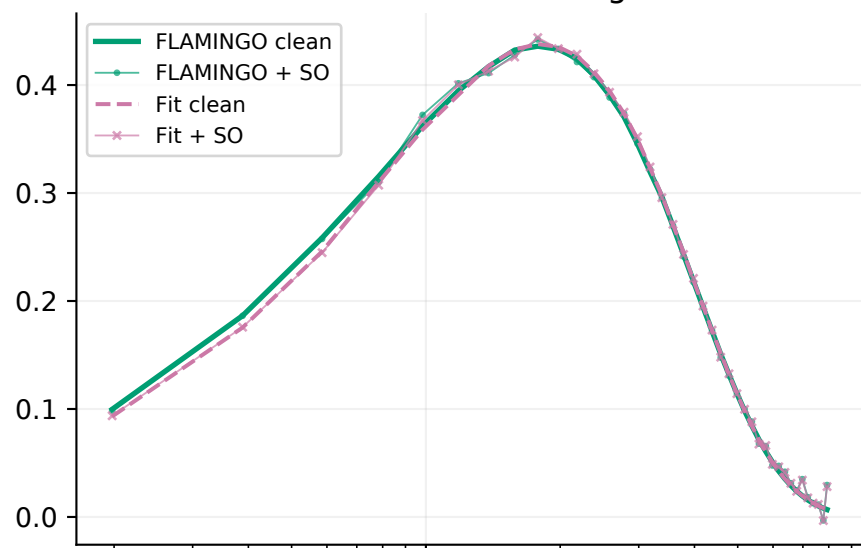
FLAMINGO fiducial



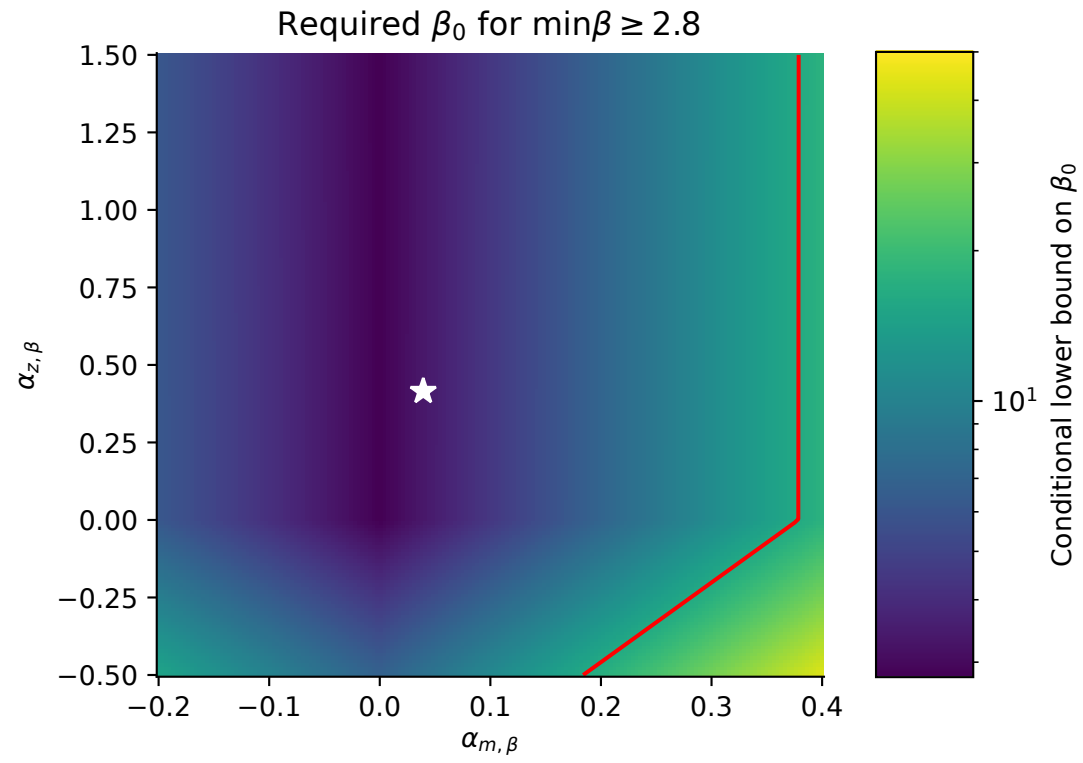
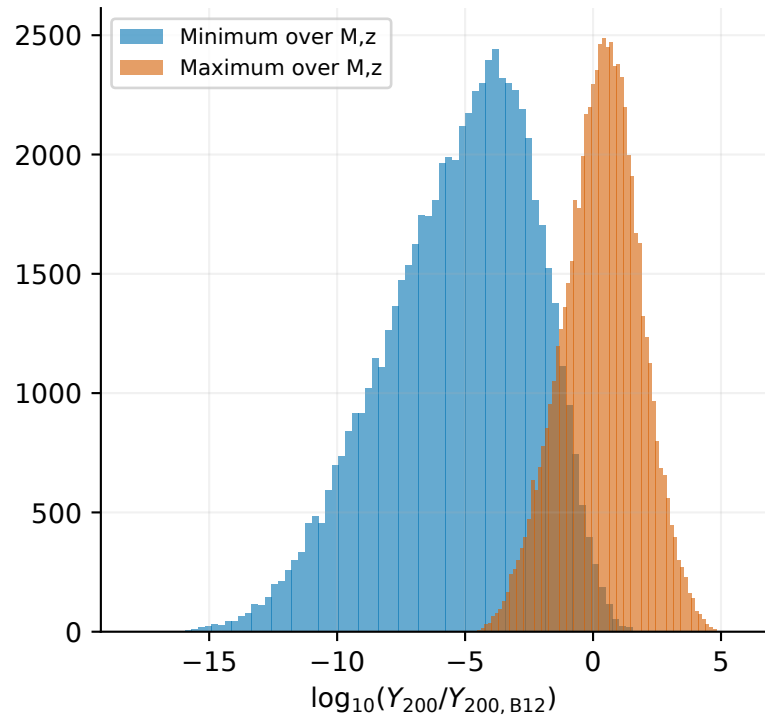
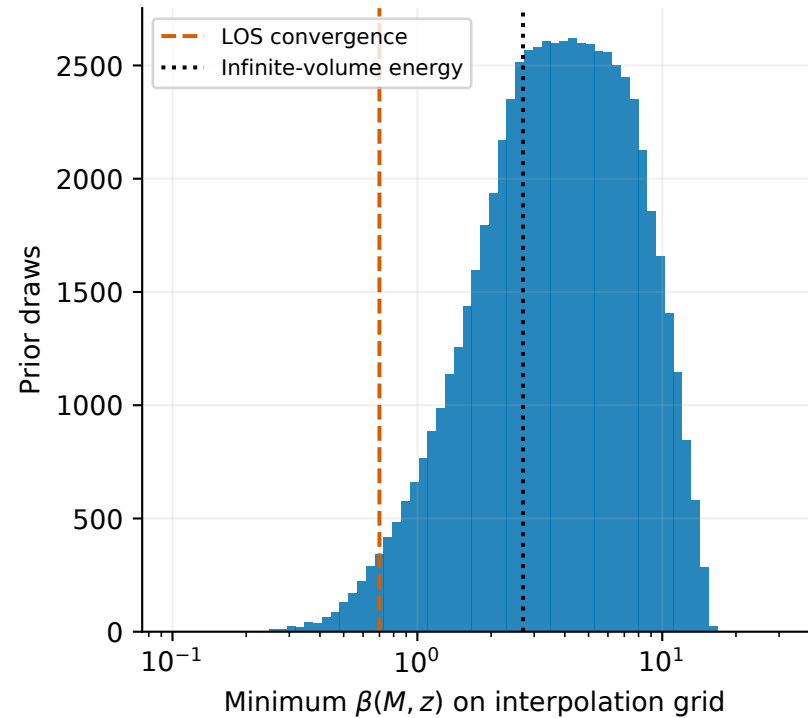
FLAMINGO fgas -8 sigma



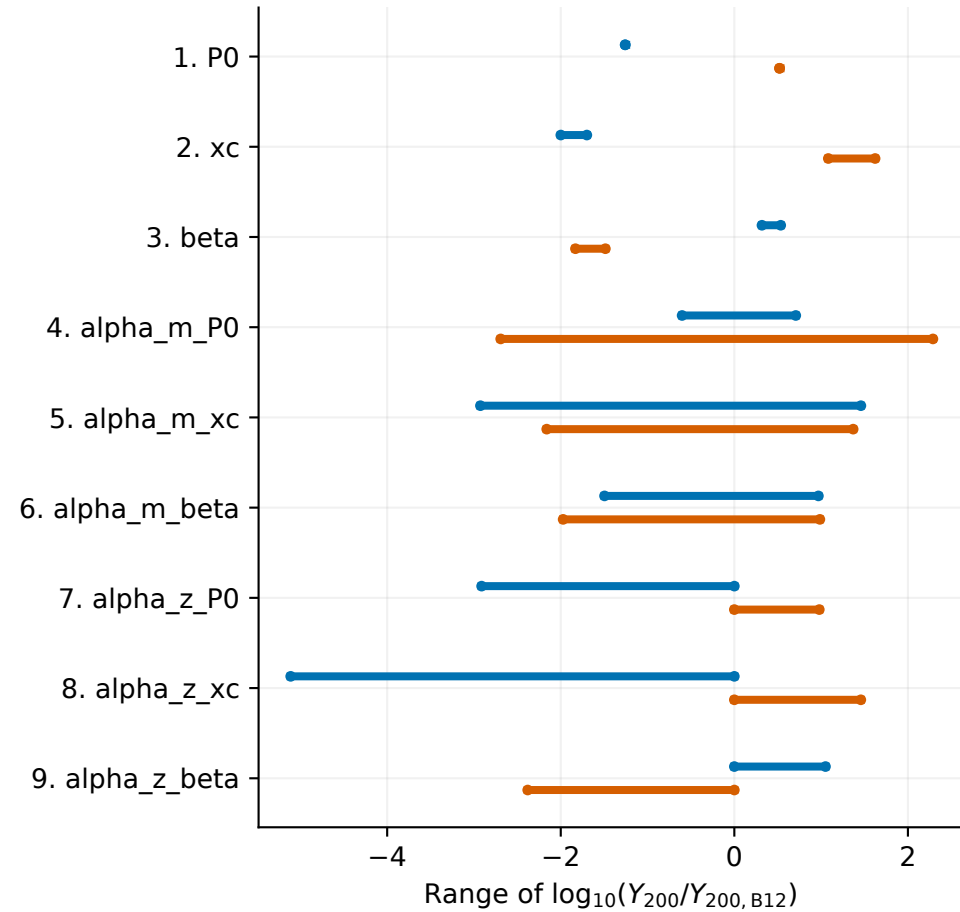
FLAMINGO Mstar -1 sigma



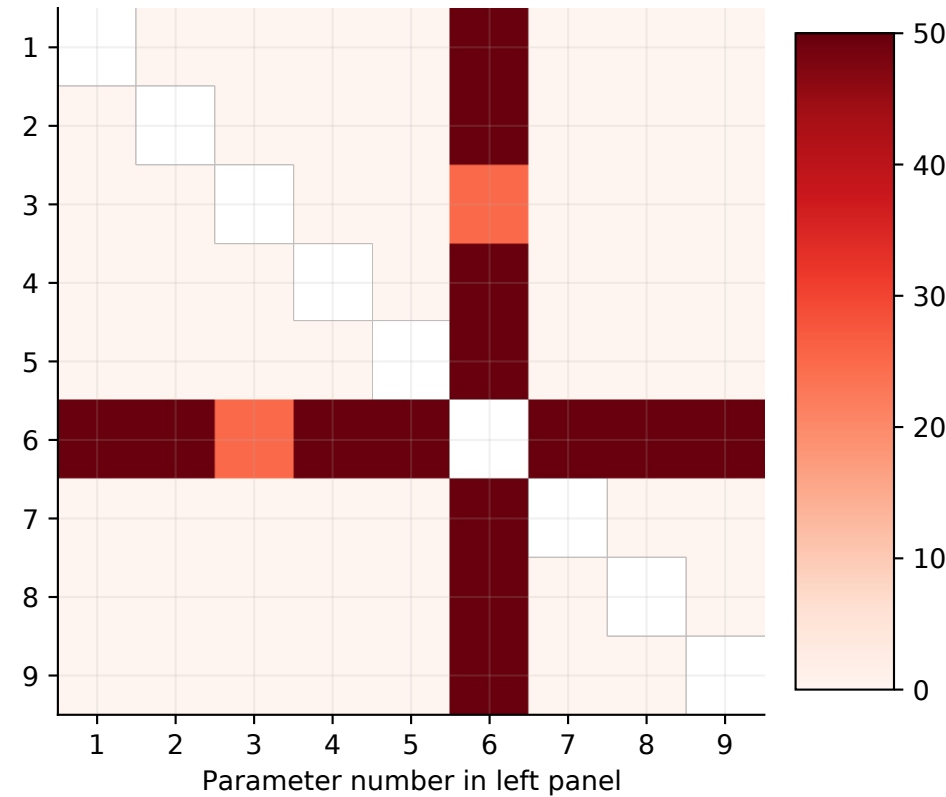
65,536 prior draws | slope grid: $1e12$ - $1e15.7$ Msun, $z=0.001$ - 5 ; Y200 grid also includes $z=0$
Red boundary: no allowed $\beta_0 \leq 16$ can satisfy the conservative outer-energy condition



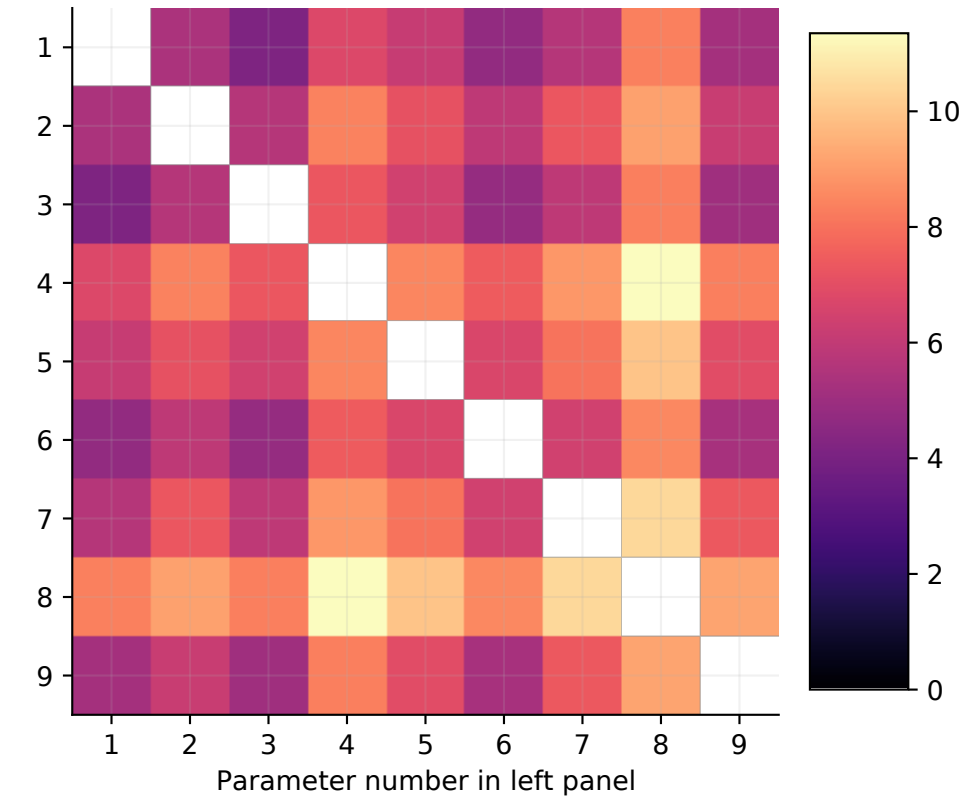
Single edges: low blue / high orange



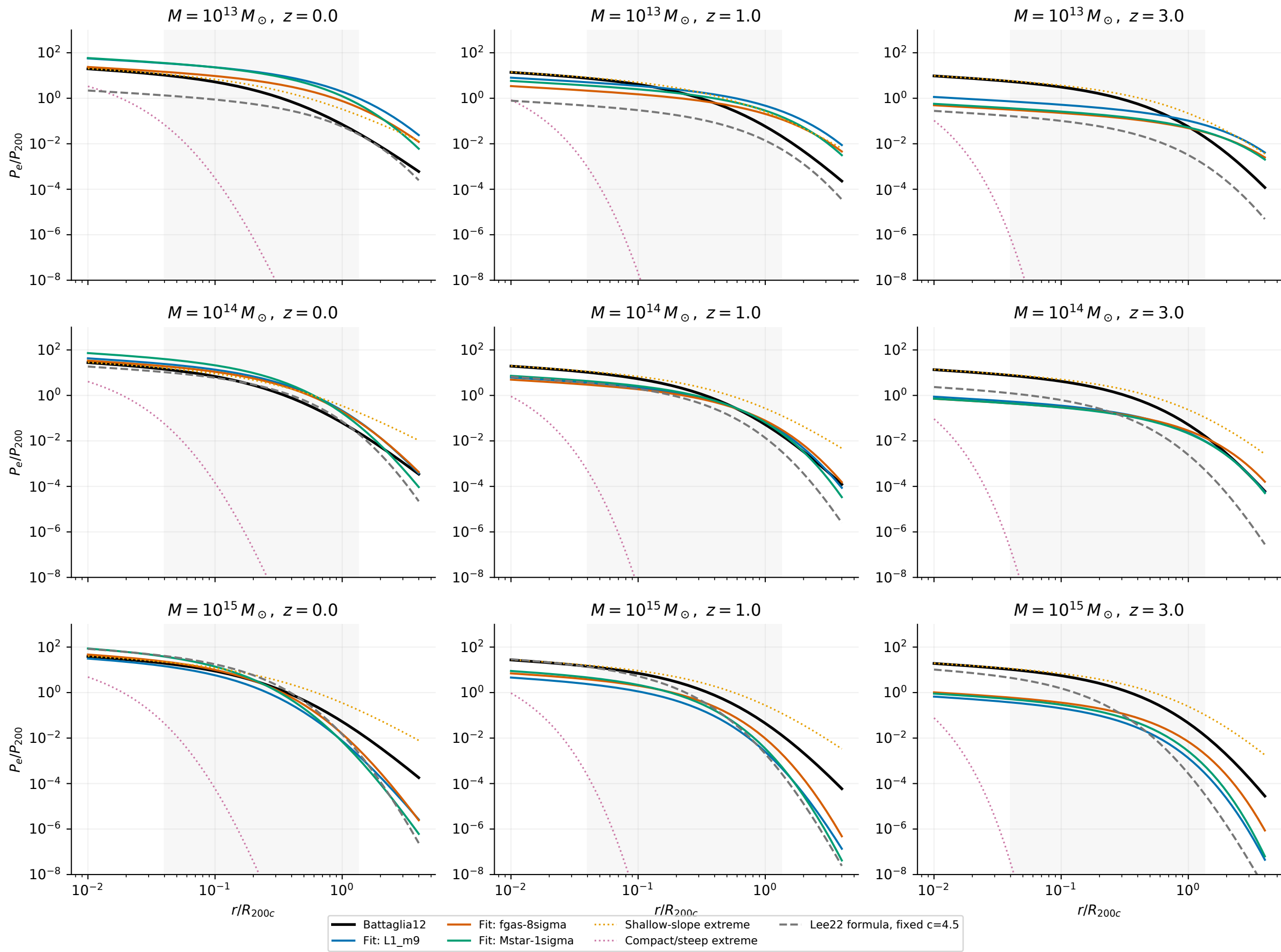
Pair corners with divergent LOS [%]

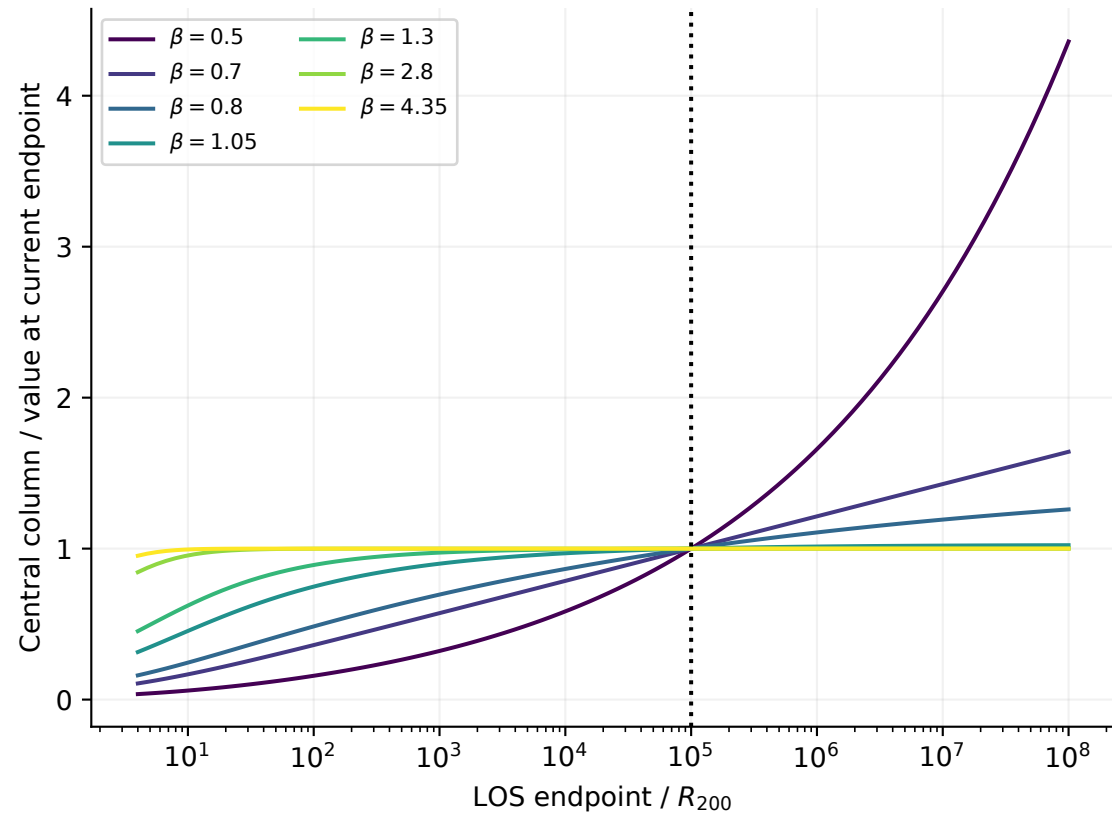


Pair-corner Y200 span [dex]

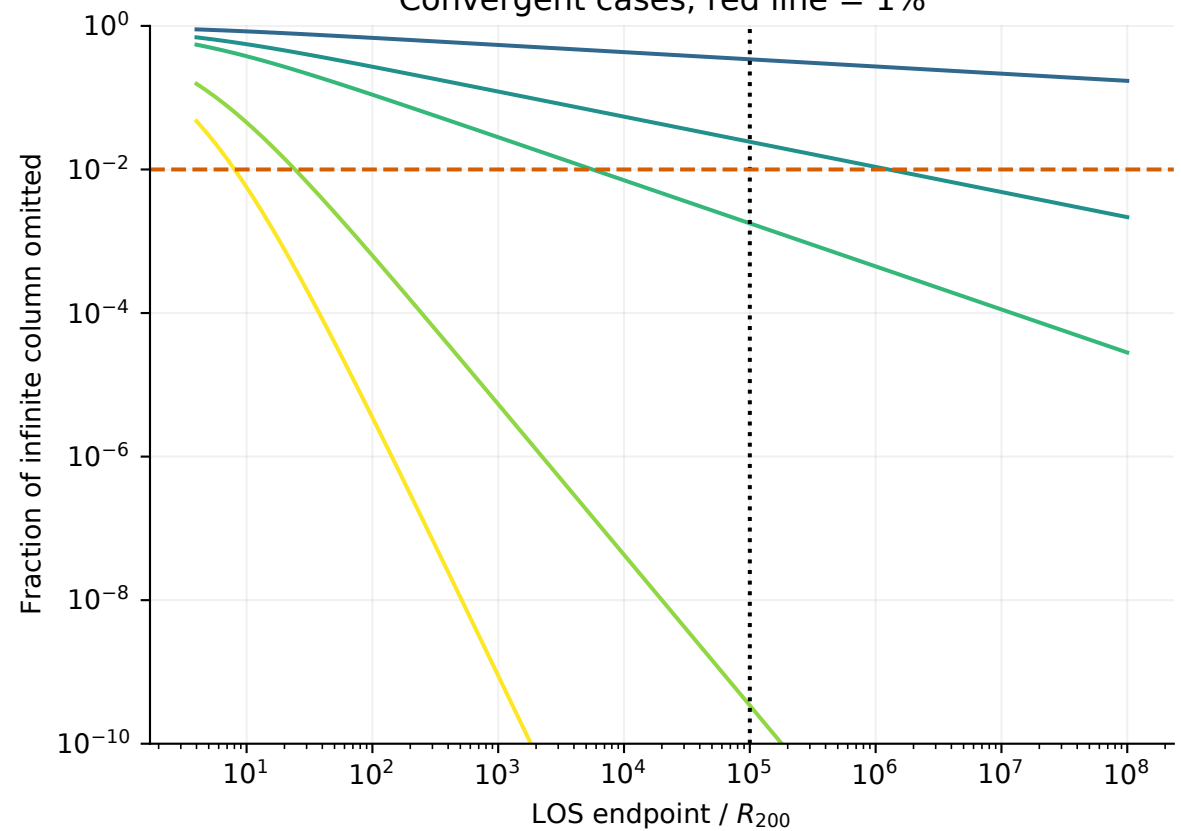


Pressure profiles implied by the spectral fits and extremes
 Grey band: Lee22 radial fit interval; mass calibration roughly $1.5e13$ – $1.5e14$ Msun, $z \leq 2$
 Other masses, redshifts and radii are illustrative extrapolations

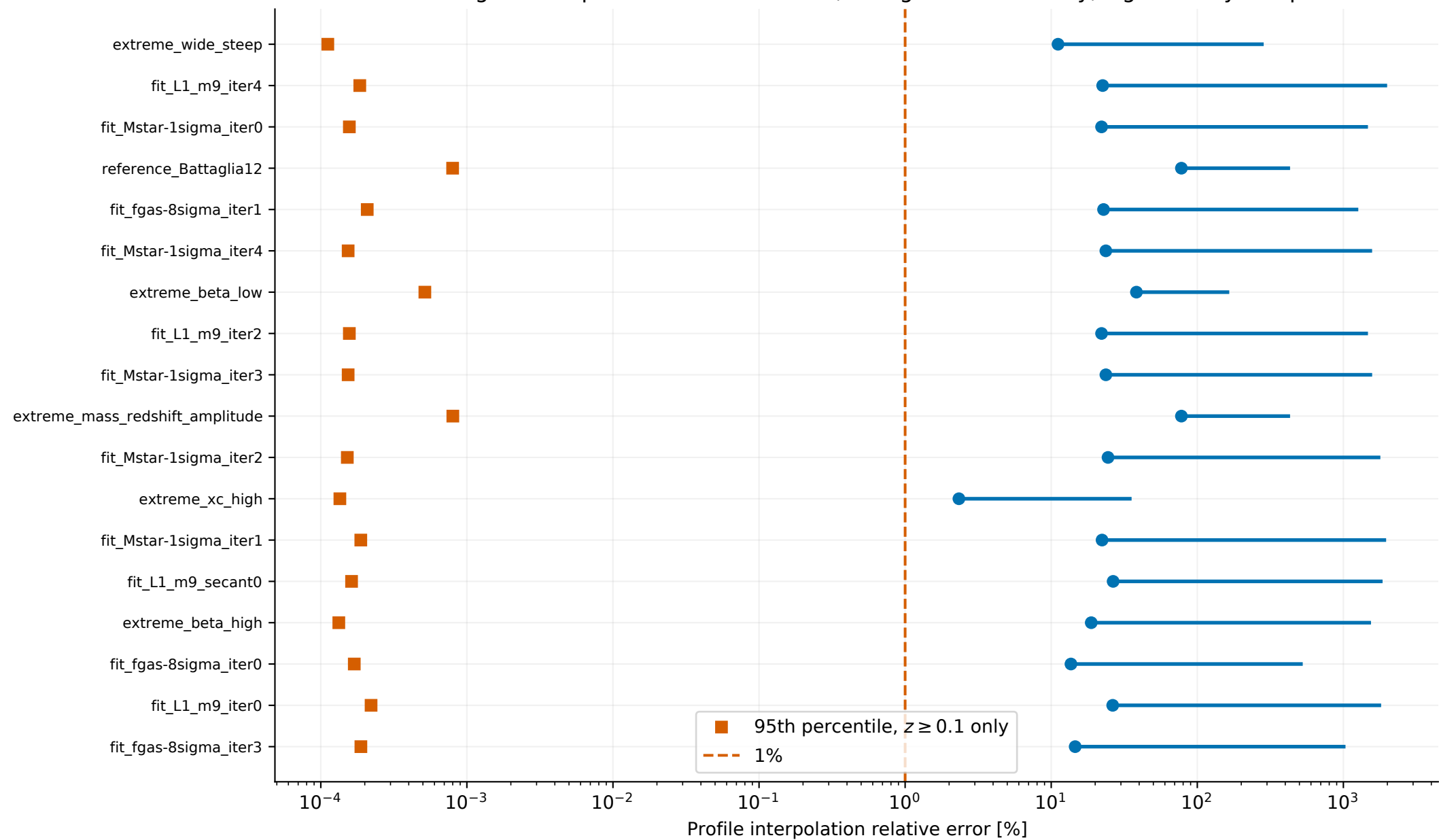


Cutoff sensitivity at $x_c = 4$ and fixed local beta

Convergent cases; red line = 1%



Independent LOS quadrature versus production interpolation
Blue: all-grid 95th percentile to maximum; orange: $z \geq 0.1$ only; significant y samples



Pixel sampling at $M=1e14$ Msun: 32 actual HEALPix placements per case
 Lines: median; bands: 16–84 percentiles; last panel logarithmic. A beam cannot restore missing flux.

